The documentation and process conversion measures necessary to comply with this revision shall be completed by 24 March 2014.

INCH-POUND

MIL-PRF-19500/463K
24 January 2014
SUPERSEDING
MIL-PRF-19500/463J
12 September 2012

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON, CURRENT REGULATOR, TYPES 1N5283-1 THROUGH 1N5314-1, AND 1N5283UR-1 THROUGH 1N5314UR-1, 1N7048-1 THROUGH 1N7055-1, 1N7048UR-1 THROUGH 1N7055UR-1, JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for 100 volt, silicon, current regulator diodes. Four levels of product assurance are provided for each encapsulated device type as specified in MIL-PRF-19500. Two levels of product assurance are provided for each unencapsulated device type.

1.2 Physical dimensions. See figure 1 (DO-7), figure 2 (DO-213AB), figure 3 (JANHCA and JANKCA), and figure 4 (JANHCB and JANKCB).

1.3 Maximum ratings. Maximum ratings are as shown in maximum test ratings (see 3.10) and as follows:

a. P_T = 500 mW (DO-7) at T_L = +50°C, L = .375 inch (9.53 mm); both ends of case or diode body to heat sink at L = .375 inch (9.53 mm). (Derate to 0 at +175°C).

b. P_T = 500 mW (DO-213AB) at T_{EC} = +125°C. (Derate to 0 at +175°C).

c. -65°C ≤ T_J ≤ +175°C; -65°C ≤ T_{STG} ≤ +175°C.

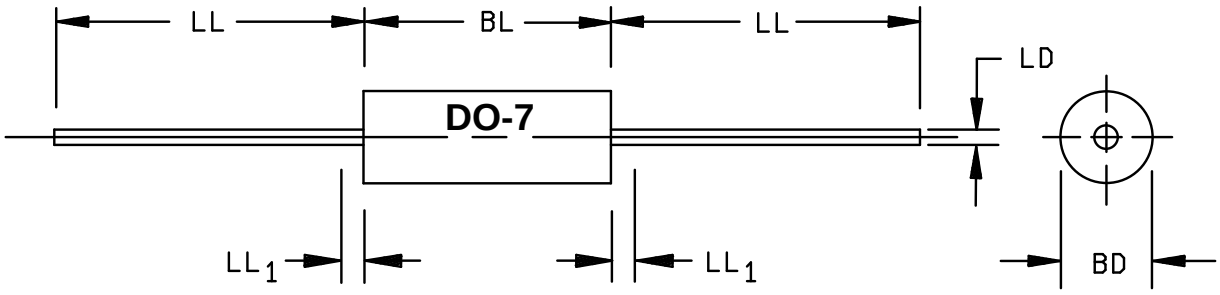
1.4 Primary electrical characteristics. Primary electrical ratings are as shown in maximum test ratings (see 3.10) and as follows, (nominally .22 mA dc ≤ I_S ≤ 4.70 mA dc, (symbol "I_P" may be used in place of symbol "I_S"):

a. $R_{\theta JL}$ = 250°C/W (maximum) at L = .375 inch (9.53 mm) (DO-7).

b. $R_{\theta JC}$ = 100°C/W (maximum) junction to end-caps (DO-213AB).

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

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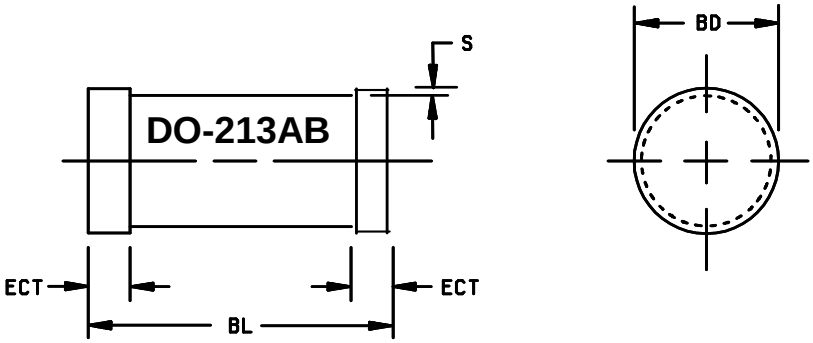


Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BD	.060	.107	1.52	2.72
BL	.120	.300	3.05	7.62
LD	.018	.023	0.46	0.58
LL	1.000	1.500	25.40	38.10
LL_1		0.050		1.27

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. The minimum body diameter shall be maintained over .15 inch (3.81 mm) inch of body length.
4. The specified lead diameter applies in the zone between .050 inch (1.27 mm) and the end of the lead. Outside of this zone the lead diameter shall not exceed LD.
5. Both leads shall be within the specified dimension.
6. See 3.3 for L and T_L definitions.
7. In accordance with ASME Y14.5M, diameters are equivalent to Φx symbology.

FIGURE 1. Physical dimensions (DO-7).

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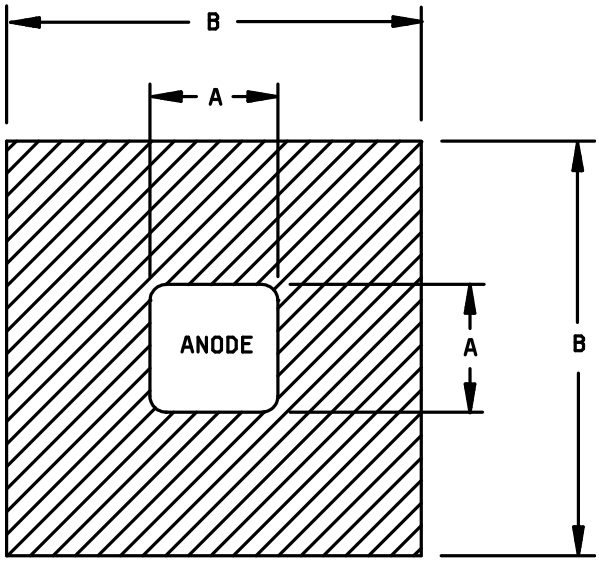


Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BD	.094	.105	2.39	2.67
BL	.189	.205	4.80	5.21
ECT	.016	.022	0.41	0.55
S	.001 min		0.03 min	

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. In accordance with ASME Y14.5M, diameters are equivalent to Φx symbology.

FIGURE 2. Physical dimensions (DO-213AB).

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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.012	.014	0.305	0.355
B	.026	.030	0.660	0.762

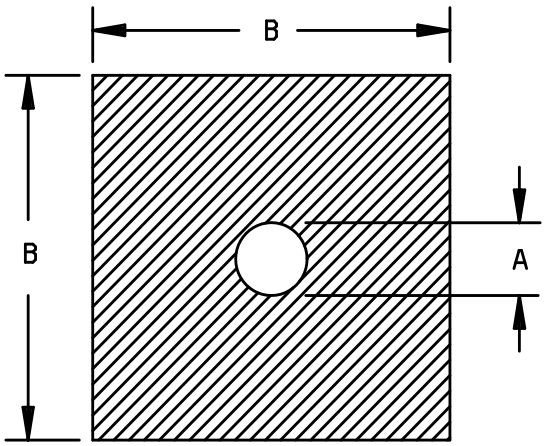
Design data

Metallization:
Top: (Anode) ..... Al
Back: (Cathode) .... Au
Al thickness ..... 25000 Å Min
Gold thickness ..... 4000 Å Min
Chip thickness ..... .010 ±.002 inch (0.254 ±0.0508 mm)

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. In accordance with ASME Y14.5M, diameters are equivalent to Φx symbology.

FIGURE 3. Physical dimensions, JANHCA and JANKCA die.

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Ltr	Inches		Millimeters	
	Min	Max	Min	Max
A	.006	.007	0.152	0.178
B	.022	.026	0.559	0.660

Design data

Metallization:
Top: (Anode) ..... Al
Back: (Cathode) .... Au
Al thickness ..... 25000 Å Min
Gold thickness ..... 4000 Å Min
Chip thickness ..... .010 ±.002 inch (0.254 ±0.0508 mm)

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. In accordance with ASME Y14.5M, diameters are equivalent to Φx symbology.

* FIGURE 4. Physical dimensions, JANHCB and JANKCB die.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at http://quicksearch.dla.mil or https://assist.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

* 2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list (QML) before contract award (see 4.2 and 6.3).

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3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions shall be as specified in MIL-PRF-19500 and as follows:

I_P	Pinch-off current. I_P Pinch-off current is defined as the regulator current at specified test voltage, V_S.
ΔI_P	Regulator current variation.
α_{IP}	Temperature coefficient of regulator current.
I_S	Pinch-off current. I_S Pinch-off current is defined as the regulator current at specified test voltage, V_S. I_S is the preferred symbol however I_P can still be used.
ΔI_S	Regulator current variation.
α_{IS}	Temperature coefficient of regulator current.
L	Lead thermal path length. Lead thermal path length is the distance from the end of the diode body to the point of lead-temperature measurement. For purposes of this measurement, the same heat sinking at the same distance from the diode body shall be applied to each lead. No heat sinking shall occur between the diode body and the point of lead-temperature measurement. This measurement may be made from either end of the diode body. (The diode body includes slugs, if any, but does not include braze fillet, paint, etc., within the zone of uncontrollable lead diameter.)
P_D	Steady-state power dissipation. Power dissipated under steady-state conditions.
T_L	Lead temperature. Lead temperature is the temperature of the lead measured at the lead thermal path length, L. Lead temperature shall be measured by means of a No. 30 copper-constantan thermocouple, or equivalent. All reference to T_L is T_{EC} for "UR" devices.
V_{POV}	Peak operating voltage. Peak operating voltage is the maximum voltage that shall be applied to the device.

* 3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, figure 1 (DO-7), figure 2 (DO-213AB), figure 3 (JANHCA and JANKC die), and figure 4 (JANKCB and JANKCB) herein.

3.5 Dash-one construction. These devices shall be of double plug construction utilizing high temperature metallurgical bonding between both sides of the silicon die and terminal pins. Metallurgical bond shall be in accordance with the requirements of category I or II in appendix A of MIL-PRF-19500.

3.5.1 JANS construction. Construction shall be dash-one, category I or II metallurgical bond in accordance with appendix A of MIL-PRF-19500.

3.5.2 Encapsulant material. In addition to those categories of hermetically sealed package requirements specified in MIL-PRF-19500, fused-metal-oxide to metal shall also be acceptable.

3.6 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.7 Marking. Marking shall be in accordance with MIL-PRF-19500. Manufacturers identification and date code shall be marked on the devices. The polarity shall be indicated with a contrasting color band to denote the cathode end. No color coding shall be permitted. Initial container package marking shall be in accordance with MIL-PRF-19500.

3.7.1 Marking of UR devices. UR devices shall be marked with a cathode band as a minimum; or a minimum of three evenly spaced contrasting color dots around the periphery of the cathode end may be used. At the option of the manufacturer, UR devices may include laser marking on an end-cap, to include part number and lot date code for all levels. JANS devices which are laser marked shall also include serialization. The prefixes JAN, JANTX, JANTXV, or JANS may be abbreviated as J, JX, JV, or JS, respectively. (For example: The part number may be reduced to JS5314). All marking which is omitted from the body of the device shall appear on the initial container. All device marking, except for polarity and serial numbers, shall also appear on the unit package used as the initial protection for delivery.

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3.8 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I and the electrical characteristics table herein.

3.9 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I herein.

3.10 Maximum test ratings. Test ratings shall be as shown in the electrical characteristics table.

3.11 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of group E tests, the tests specified in 4.7.4 herein shall be performed on the first inspection lot to this revision to maintain qualification.

4.2.2 JANHC and JANKC devices. Qualification for shall be in accordance with appendix G of MIL-PRF-19500.

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4.3 Screening (JAN, JANTXV, JANTX, and JANS levels). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table E-IV of MIL-PRF-19500)	Measurement		
	JANS	JANTX and JANTXV levels	JAN level (1)
3a	Temperature cycling	Temperature cycling	Temperature cycling (in accordance with MIL-PRF-19500, JANTX level)
(2) 3c	Thermal impedance (see 4.3.3)	Thermal impedance (see 4.3.3)	Thermal impedance (see 4.3.3)
9	I_{S1} (3)	Not applicable	Not applicable
10	V_{POV} = Col 11, table II at T_A = +25°C t = 48 hours	V_{POV} = Col 11, table II at T_A = +25°C t = 48 hours	V_{POV} = Col 11, table II at T_A = +25°C t = 48 hours
11	Subgroup 2 of table I herein; $\Delta I_{S1} \leq$ 5 percent of initial value (4)	Subgroup 2 of table I herein	Subgroup 2 of table I herein
12	See 4.3.2	See 4.3.2	Not applicable
(5) 13	Subgroup 2 of table I herein; $\Delta I_{S1} \leq$ 5 percent of initial value. (4)	Subgroup 2 of table I herein; $\Delta I_{S1} \leq$ 5 percent of initial value. (4)	Not applicable

(1) Screens 3a, 3c, 10, and 11 are the only screens required for JAN level product.
(2) Shall be performed any time after temperature cycling, screen 3a; JANTX and JANTXV levels do not need to be repeated in screening requirements.
(3) Symbol "I_{P1}" may be used in place of "I_{S1}".
(4) Symbol "ΔI_{P1}" may be used in place of "ΔI_{S1}".
(5) When thermal impedance is performed prior to screen 13, it is not required to be repeated in screen 13.

4.3.1 Screening (JANHC or JANKC). Screening of die shall be in accordance with appendix G of MIL-PRF-19500. As a minimum, die shall be 100-percent probed to ensure compliance with table I, subgroup 2 (with the exception of thermal impedance).

4.3.2 Power burn-in conditions. Power burn-in conditions are as follows: I_R = 200 mA dc minimum; mounting and test conditions in accordance with method 1038 of MIL-STD-750, test condition B, T_{EC} = +75°C to +125°C for surface mount devices. T_A = room ambient as defined in the general requirements of 4.5 of MIL-STD-750.

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4.3.3 Thermal impedance $Z_{\theta JX}$ measurements for screening. The $Z_{\theta JX}$ measurements shall be performed in accordance with method 3101 of MIL-STD-750, (V_R to be used in lieu of V_F). The maximum limit (not to exceed the table I, subgroup 2 limit) for $Z_{\theta JX}$ in screening (table E-IV of MIL-PRF-19500) shall be derived by each vendor bymeans of statistical process control. When the process has exhibited control and capability, the capability data shall be used to establish the fixed screening limit. In addition to screening, once a fixed limit has been established, monitor all future sealing lots using a random five piece sample from each lot to be plotted on the applicable X, R chart. If a lot exhibits an out of control condition, the entire lot shall be removed from the line and held for engineering evaluation and disposition.

a.	I_M measurement current	1 mA - 10 mA.
b.	I_H forward heating current	.5 A - 1.0 A.
c.	t_H heating time	10 ms.
d.	t_{MD} measurement delay time	70 μs maximum.

4.3.3.1 For initial qualification or requalification. Read and record data ($Z_{\theta JX}$) shall be supplied to the qualifying activity on one lot (random sample of 500 devices minimum) prior to shipment. Twenty-two samples shall be serialized and provided to the qualifying activity for test correlation.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in tables E-VIA (JANS) and E-VIB (JAN, JANTX and JANTXV) of MIL-PRF-19500, and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with table I, subgroup 2 herein.

4.4.2.1 Group B inspection, table E-VIA (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B4	1037	2,000 cycles; test conditions in accordance with 4.3.2: $t_{on} = t_{off}$ 30 seconds minimum.
B5	1027	I_R = 200 mA dc, T_A = +125°C or adjusted as required to give an average lot T_J = +175°C. Marking legibility requirements shall not apply.

4.4.2.2 Group B inspection, table E- VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
B3	1027	V_{POV} = Col 11, table II; T_A = +25°C; L = .375 inch (9.53 mm) (non-surface mount), L = 0 inch for surface mount.
B5		Not applicable.
B6	1032	T_A = +175°C.

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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with table I, subgroup 2 herein.

Subgroup	Method	Condition
C2	2036	Tension - test condition A; weight = 10 pounds (4.54 Kg), t = 15 s; lead fatigue = condition E (not applicable to 'UR' suffix types).
C5	3101 or 4081	$R_{\theta JL}$ at L = .375 inch (9.52 mm) ≤ 250°C/W, $R_{\theta JEC}$ at L = 0 lead length ≤ 100°C/W, see 4.5. $R_{\theta JEC}$ = 100°C/W (maximum) at zero lead length (for UR).
C6	1026	V_{POV} = Col 11, table II; T_A = +25°C; L = .375 inch (9.53 mm) (non-surface mount), L = 0 inch for surface mount.
C7		See 4.6 and 4.7.3.

4.5 Thermal resistance. Thermal resistance measurement shall be in accordance with method 3101 of MIL-STD-750. Forced moving air or draft shall not be permitted across the device during heat. The maximum limit for $R_{\theta JL}$ under these test conditions shall be $R_{\theta JL}$ (max) = 250°C/W or $R_{\theta JEC}$ = 100°C/W. The following conditions shall apply:

a. I_H 200 mA to 400 mA .

b. I_M 1 mA to 10 mA.

c. t_H 30 seconds minimum.

d. t_{MD} 70 μs maximum.

4.5.1 For initial qualifications and re-qualifications. Read and record data in accordance with 4.7.4 herein and shall be included in the qualification report.

4.6 Temperature coefficient of regulator current. The temperature coefficient of regulator current shall be tested under the following conditions: Sampling plan: 22 devices, c = 0.

a. Test 1: V_S = 25 V dc, T_{L1} = -55°C, T_{L2} = +25°C, L = .375 inch (9.53 mm) (non-surface mount), L = 0 inch (surface mount) (see 3.3 and 4.7.3) with the maximum limit in accordance with column 8 of table II herein.

b. Test 2: V_S = 25 V dc, T_{L1} = +25°C, T_{L2} = +150°C, L = .375 inch (9.53 mm) (non-surface mount), L = 0 inch (surface mount) (see 3.3 and 4.7.3) with the maximum limit in accordance with column 9 of table II herein.

4.7 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.7.1 Regulator impedance (Z_S) at test voltage V_S. To test for Z_S, a 90 Hz signal V_S (mod) with rms value equal to 10 percent of test voltage, V_S, is superimposed on the test voltage (see the regulator impedance test circuit figure herein).

4.7.2 Knee ac impedance (Z_K) at test voltage V_K. To test for Z_K, a 90 Hz signal V_K (mod) with rms value equal to 10 percent of test voltage, V_K, is superimposed on the test voltage (see the knee impedance test circuit figure herein).

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4.7.3 Temperature coefficient of regulator current (α_{IS}). Temperature coefficient of regulator current shall be calculated as follows:

$$\alpha_{IS} = \frac{I_S (T_{L2}) - I_S (T_{L1})}{I_S (T_L = +25°C)\Delta T_L} \times 100$$

4.7.3.1 Alternate definition of temperature coefficient of regulator current (α_{IS}). Temperature coefficient of regulator current is allowed to also be calculated as follows:

$$\alpha_{IS} = \frac{I_P (T_{L2}) - I_P (T_{L1})}{I_P (T_L = +25°C)\Delta T_L} \times 100$$

4.7.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps and footnotes of table I, subgroup 2 herein.

4.7.4.1 Group E inspection, table E- IX of MIL-PRF-19500.

Inspections	MIL-STD-750		Sample plan
	Method	Condition	
E1	1051	500 cycles.	45 devices, c = 0
E2	1037	6,000 cycles (see 4.3.2) $t_{on} = t_{off} = 30$ seconds minimum.	45 devices, c = 0
E3	2101	Cross section; scribe and break. Separate samples to be used for each test.	3 devices, c = 0
E4		See table E-IX of MIL-PRF-19500, subgroup 4.	
E6	1020	As applicable.	3 devices, c = 0

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit 2/		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Regulator current		V_S = 25 V dc, t = 90 s or thermal equilibrium, 1N5283-1 through 1N5314-1 t = pulse measurement, 10 ms max 1N7048-1 through 1N7055-1, T_L = +30°C ±3°C (see figure 5)	I_{S1} 3/	Column 3	Column 4	mA dc
Limiting voltage		I_L = .8 I_{s1} (min) 3/, column 3 of table II (see figure 6)	V_L		Column 7	V dc
Reverse voltage		I_R = 200 mA	V_R		2.5	V dc
Thermal impedance	3101	See 4.3.3	$Z_{\theta JX}$		25	°C/W
Subgroup 3						
Not applicable						
Subgroup 4						
Regulator impedance		V_S = 25 V dc; (see figure 7 and 4.7.1)	Z_S	Column 5		MΩ
Knee impedance		V_K = 6.0 V dc, (see figure 8 and 4.7.2)	Z_K	Column 6		MΩ
Subgroups 5 and 6						
Not applicable						
Subgroup 7						
Regulator current		V_S = column 11 of table II, t = 90 s or thermal equilibrium, 1N5283-1 through 1N5314-1 t = pulse measurement, 10 ms max 1N7048-1 through 1N7055-1, T_L = +30°C ±3°C (see figure 5)	I_{S2} 4/		Column 10	mA dc

1/ For sampling plan, see MIL-PRF-19500.
2/ Column references are to table II herein.
3/ Symbol "I_{P1}" may be used in place of "I_{S1}".
4/ Symbol "I_{P2}" may be used in place of "I_{S2}".

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TABLE II. Electrical characteristics. 1/

Col 1	Col 2	Col 3	Col 4	Col 5	Col 6	Col 7	Col 8		Col 9		Col 10	Col 11
Type (Electrical characteristics for "UR" and "-1" suffix devices are identical.)	I_{S1} regulator current (mA) at V_S = 25 V 2/			Z_S minimum regulator impedance at V_S = 25 V	Z_k minimum knee impedance at V_K = 6 V	V_L maximum limiting voltage at I_L = 0.8 I_{S1} (min) 2/	α_{IS} maximum regulator current T_C at V_S = 25V		α_{IS} maximum regulator current T_C at V_S = 25V		I_{S2} regulator current (mA) at V_S = Col 11 3/	V_{POV} peak operating volts (DC)
							-55°C	+25°C	+25°C	+150°C		
							(%/°C)		(%/°C)			
	Nom	Min	Max	MΩ	MΩ	Volts	Min	Max	Min	Max	Max	Volts
1N5283-1	0.22	0.198	0.242	25.0	2.75	1.00	-.20	1.15	-.16	0.60	.27	100
1N5284-1	0.24	0.216	0.264	19.0	2.35	1.00	-.20	1.05	-.20	0.56	.30	100
1N5285-1	0.27	0.243	0.297	14.0	1.95	1.00	-.30	0.95	-.22	0.48	.33	100
1N5286-1	0.30	0.270	0.330	9.0	1.60	1.00	-.35	0.85	-.25	0.42	.36	100
1N5287-1	0.33	0.297	0.363	6.6	1.35	1.00	-.40	0.75	-.26	0.37	.40	100
1N5288-1	0.39	0.351	0.429	4.10	1.00	1.05	-.50	0.62	-.30	0.28	.47	100
1N5289-1	0.43	0.387	0.473	3.30	0.870	1.05	-.52	0.55	-.32	0.23	.52	100
1N5290-1	0.47	0.423	0.517	2.70	0.750	1.05	-.55	0.50	-.33	0.18	.57	100
1N5291-1	0.56	0.504	0.616	1.90	0.560	1.10	-.60	0.35	-.36	0.10	.68	100
1N5292-1	0.62	0.558	0.682	1.55	0.470	1.13	-.62	0.25	-.37	0.05	.75	100
1N5293-1	0.68	0.612	0.748	1.35	0.400	1.15	-.65	0.20	-.38	0.02	.82	100
1N5294-1	0.75	0.675	0.825	1.15	0.335	1.20	-.70	0.15	-.40	-.03	.91	100
1N5295-1	0.82	0.738	0.902	1.00	0.290	1.25	-.72	0.07	-.41	-.07	.99	100
1N5296-1	0.91	0.819	1.001	0.880	0.240	1.29	-.76	0.0	-.42	-.10	1.10	100
1N5297-1	1.00	0.900	1.100	0.800	0.205	1.35	-.78	0.05	-.44	-.10	1.21	100
1N5298-1	1.10	0.990	1.210	0.700	0.180	1.40	-.80	-.10	-.46	-.10	1.33	100
1N5299-1	1.20	1.08	1.32	0.640	0.155	1.45	-.83	-.15	-.47	-.10	1.45	100
1N5300-1	1.30	1.17	1.43	0.580	0.135	1.50	-.85	-.20	-.48	-.10	1.57	100
1N5301-1	1.40	1.26	1.54	0.540	0.115	1.55	-.88	-.20	-.49	-.10	1.69	100
1N5302-1	1.50	1.35	1.65	0.510	0.105	1.60	-.90	-.20	-.50	-.10	1.81	100

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TABLE II. Electrical characteristics - Continued. 1/

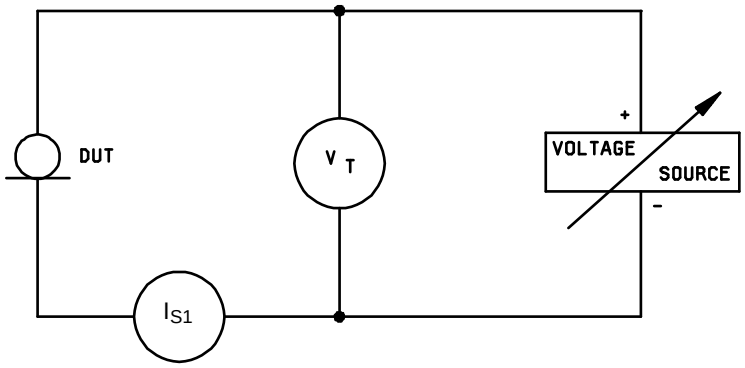
Col 1	Col 2	Col 3	Col 4	Col 5	Col 6	Col 7	Col 8		Col 9		Col 10	Col 11
Type (Electrical characteristics for "UR" and "-1" suffix devices are identical)	I_{S1} regulator current (mA) at V_S = 25 V 2/			Z_S minimum regulator impedance at V_S = 25 V	Z_k minimum knee impedance at V_K = 6 V	V_L maximum limiting voltage at I_L = 0.8 I_{S1} (min) 2/	α_{IS} maximum regulator current T_C at V_S = 25V		α_{IS} maximum regulator current T_C at V_S = 25V		I_{S2} regulator current (mA) at V_S = Col 11 3/	V_{POV} peak operating volts (DC)
							-55°C	+25°C	+25°C	+150°C		
							(%/°C)		(%/°C)			
	Nom	Min	Max	MΩ	MΩ	Volts	Min	Max	Min	Max	Max	Volts
1N5303-1	1.60	1.44	1.76	0.475	0.092	1.65	-.90	-.20	-.50	-.10	1.92	100
1N5304-1	1.80	1.62	1.98	0.420	0.074	1.75	-.92	-.20	-.51	-.10	2.18	100
1N5305-1	2.00	1.80	2.20	0.395	0.061	1.85	-.95	-.20	-.52	-.10	2.42	100
1N5306-1	2.20	1.98	2.42	0.370	0.052	1.95	-.96	-.20	-.52	-.10	2.66	100
1N5307-1	2.40	2.16	2.64	0.345	0.044	2.00	-.98	-.20	-.53	-.10	2.90	100
1N5308-1	2.70	2.43	2.97	0.320	0.035	2.15	-1.0	-.20	-.53	-.10	3.27	100
1N5309-1	3.00	2.70	3.30	0.300	0.029	2.25	-1.01	-.20	-.53	-.10	3.63	100
1N5310-1	3.30	2.97	3.63	0.280	0.024	2.35	-1.02	-.20	-.54	-.10	3.99	100
1N5311-1	3.60	3.24	3.96	0.265	0.020	2.50	-1.03	-.20	-.54	-.10	4.36	100
1N5312-1	3.90	3.51	4.29	0.255	0.017	2.60	-1.04	-.20	-.55	-.10	4.72	100
1N5313-1	4.30	3.87	4.73	0.245	0.014	2.75	-1.05	-.20	-.55	-.10	5.20	100
1N5314-1	4.70	4.23	5.17	0.235	0.012	2.90	-1.06	-.20	-.55	-.10	5.69	100
1N7048-1	5.1	4.59	5.61	0.100	0.004	3.67	-1.06	-.20	-.55	-.10	6.89	80
1N7049-1	5.6	5.04	6.16	0.090	0.004	4.03	-1.06	-.20	-.55	-.10	7.54	80
1N7050-1	6.2	5.58	6.82	0.080	0.003	4.46	-1.06	-.20	-.55	-.10	8.38	70
1N7051-1	6.8	6.12	7.48	0.070	0.002	4.90	-1.06	-.20	-.55	-.10	9.20	70
1N7052-1	7.5	6.75	8.25	0.050	0.0015	5.40	-1.06	-.20	-.55	-.10	10.20	60
1N7053-1	8.2	7.38	9.02	0.030	0.0015	5.90	-1.06	-.20	-.55	-.10	11.20	60
1N7054-1	9.1	8.19	10.01	0.020	0.001	6.55	-1.06	-.20	-.55	-.10	12.40	50
1N7055-1	10.0	9.00	11.10	0.010	0.001	7.20	-1.06	-.20	-.55	-.10	14.40	50

1/ Electrical characteristics are for all package styles.

2/ Symbol "I_{P1}" may be used in place of "I_{S1}".

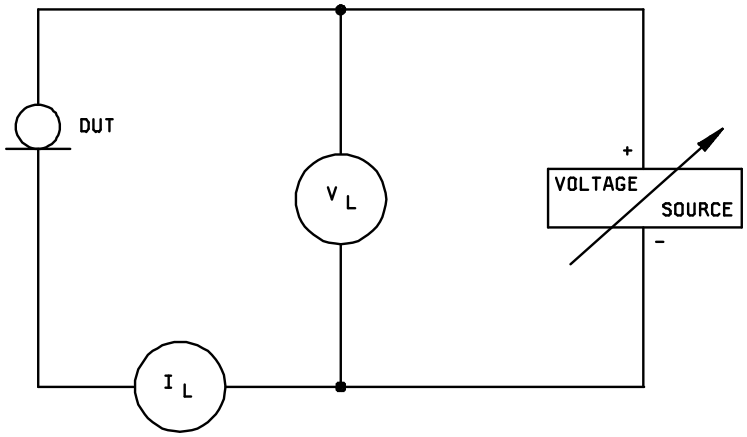
3/ Symbol "I_{P2}" may be used in place of "I_{S2}".

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NOTES:
1. Adjust voltage source so that V_S = 25 V dc.
2. Measure current I_{S1}. Symbol "I_{P1}" may be used in place of "I_{S1}".
3. The device is acceptable if the current falls within the limits specified.
4. The ammeter shall represent essentially a short-circuit to the terminals between which the current is being measured. If not, the voltmeter reading shall be corrected for the drop across the ammeter.

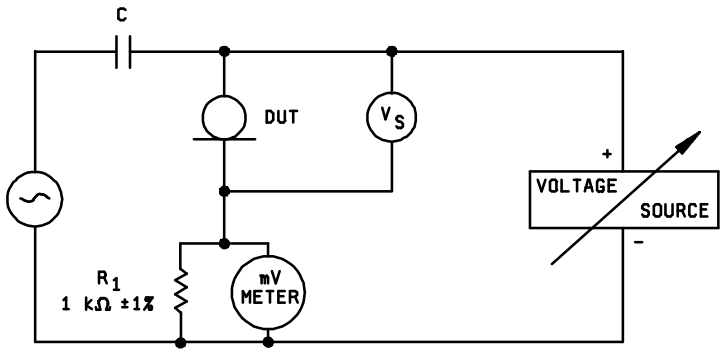
FIGURE 5. Regulator current test circuit.



NOTES:
1. Adjust current source so that I_L = .8 I_{S1} (min). Symbol "I_{P1}" may be used in place of "I_{S1}".
2. Measure voltage V_L.
3. The device is acceptable if the voltage is less than the limit specified.
4. The ammeter shall represent essentially a short-circuit to the terminals between which the current is being measured. If not, the voltmeter reading shall be corrected for the drop across the ammeter.

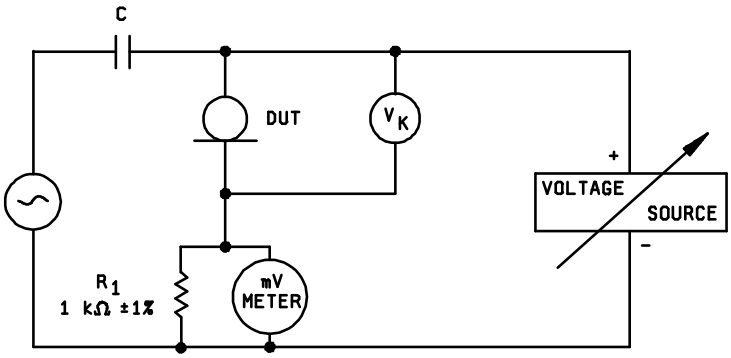
FIGURE 6. Limiting voltage test circuit.

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NOTES:
1. Adjust voltage source so that V_S = 25 V dc.
2. Apply an ac signal of 2.5 V rms at 90 Hz through an isolating capacitor C.
3. Measure the ac rms voltage.
4. $Z_S = V_S$ mod x (R1 ÷ V ac) where V_S mod equals ac signal for note 2 and V ac equals the voltage across R1.
5. Device is acceptable if the regulator impedance meets the specified minimum limit.

FIGURE 7. Regulator impedance test circuit.



NOTES:
1. Adjust voltage source so that V_K = 6.0 V dc.
2. Apply an ac signal of .6 Vrms at 90 Hz through an isolating capacitor C.
3. Measure the ac rms voltage.
4. $Z_K = V_K$ mod x (R_1 ÷ V ac) where V_K mod equals ac signal for note 2 and V ac equals the voltage across R_1. Device is acceptable if the knee impedance meets the specified minimum limit.

FIGURE 8. Knee impedance test circuit.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors devices conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.6).

d. Product assurance level and type designator.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

* 6.4 Suppliers of die. The qualified die suppliers with the applicable letter version (example JANHCA1N5283) will be identified on the QML.

JANC ordering information		
PIN	Manufacturer CAGE	
1N5283-1 through 1N5314-1	43611	52GC4
	JANHCA1N5283 through JANHCA1N5314 or JANKCA1N5283 through JANKCA1N5314	JANHCB1N5283 through JANHCB1N5314 or JANKCB1N5283 through JANKCB1N5314
1N7048-1 through 1N7055-1	JANHCA1N7048 through JANHCA1N7055 or JANKCA1N7048 through JANKCA1N7055	JANHCB1N7048 through JANHCB1N7055 or JANKCB1N7048 through JANKCB1N7055

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6.5 Substitutability of dash-one parts. Non-dash-one devices have been deleted from this specification. Dash-one devices are a direct substitute for non dash-one devices and are preferred.

Superseded part number	Superseding part number		Superseded part number	Superseding part number		Superseded part number	Superseding part number
1N5283	1N5283-1		1N5294	1N5294-1		1N5305	1N5305-1
1N5283UR	1N5283UR-1		1N5294UR	1N5294UR-1		1N5305UR	1N5305UR-1
1N5284	1N5284-1		1N5295	1N5295-1		1N5306	1N5306-1
1N5284UR	1N5284UR-1		1N5295UR	1N5295UR-1		1N5306UR	1N5306UR-1
1N5285	1N5285-1		1N5296	1N5296-1		1N5307	1N5307-1
1N5285UR	1N5285UR-1		1N5296UR	1N5296UR-1		1N5307UR	1N5307UR-1
1N5286	1N5286-1		1N5297	1N5297-1		1N5308	1N5308-1
1N5286UR	1N5286UR-1		1N5297UR	1N5297UR-1		1N5308UR	1N5308UR-1
1N5287	1N5287-1		1N5298	1N5298-1		1N5309	1N5309-1
1N5287UR	1N5287UR-1		1N5298UR	1N5298UR-1		1N5309UR	1N5309UR-1
1N5288	1N5288-1		1N5299	1N5299-1		1N5310	1N5310-1
1N5288UR	1N5288UR-1		1N5299UR	1N5299UR-1		1N5310UR	1N5310UR-1
1N5289	1N5289-1		1N5300	1N5300-1		1N5311	1N5311-1
1N5289UR	1N5289UR-1		1N5300UR	1N5300UR-1		1N5311UR	1N5311UR-1
1N5290	1N5290-1		1N5301	1N5301-1		1N5312	1N5312-1
1N5290UR	1N5290UR-1		1N5301UR	1N5301UR-1		1N5312UR	1N5312UR-1
1N5291	1N5291-1		1N5302	1N5302-1		1N5313	1N5313-1
1N5291UR	1N5291UR-1		1N5302UR	1N5302UR-1		1N5313UR	1N5313UR-1
1N5292	1N5292-1		1N5303	1N5303-1		1N5314	1N5314-1
1N5292UR	1N5292UR-1		1N5303UR	1N5303UR-1		1N5314UR	1N5314UR-1
1N5293	1N5293-1		1N5304	1N5304-1			
1N5293UR	1N5293UR-1		1N5304UR	1N5304UR-1			

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Army - CR
Navy - EC
Air Force - 85
NASA - NA
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, MC
Air Force - 19, 99

Preparing activity:
DLA - CC

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